

Cryptographic Tools

-for Confidentiality and Integrity

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Cryptography in the News: The Post-Quantum Migration

- In 2026, a White House executive order set hard deadlines for federal agencies to retire RSA and ECC and migrate to "post-quantum cryptography"
- The EU, UK, Australia and others have published their own migration timelines (most targeting 2030–2035)

Question: Why is the whole world racing to replace encryption algorithms we've trusted for 40+ years before quantum computers even arrive?

DS DefenseScoop

[DOD probes industry about software-defined encryption as U.S. scurries to get quantum-safe](#)

The department is also looking into data protection measures that are designed to advance its migration to post-quantum cryptography.

2 days ago

TV TradingView

[AVGO: Quantum computing accelerates the need for urgent, risk-based migration to post-quantum cryptography](#)

Quantum computing threatens current cryptographic systems, with state-sponsored actors harvesting encrypted data for future decryption.

3 days ago

T Tech Times

[Banks Can Finish Post-Quantum Migration and Still Be Exposed: Treasury Targets Vendors](#)

Post-quantum cryptography financial sector readiness hinges on vendor migration: Treasury's Quantum-Readiness Task Force, launched August 24...

4 days ago



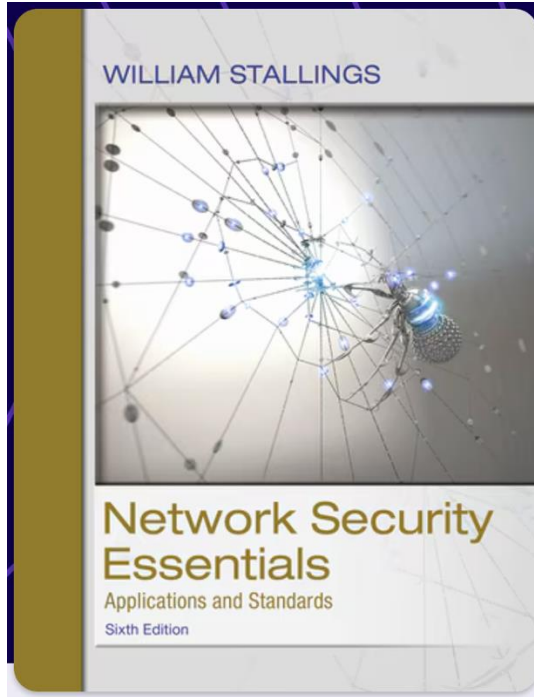
Harvest Now, Decrypt Later

- Encrypted traffic can be intercepted and stored today, then decrypted years later when the technology catches up
- **Question:** if a message encrypted today is decrypted in 10 years, does that matter? For which kinds of data?
- **Key idea:** encryption needs to remain secure for the lifetime of the data, not just the moment of transmission

Two Kinds of Cryptography, Two Different Risks

- **Shor's algorithm (quantum)** breaks RSA and ECC completely
- **Grover's algorithm (quantum)** only halves the effective strength of AES — longer keys compensate
- RSA/ECC are asymmetric; AES is symmetric — the difference in risk comes from their different mathematical foundations
- **Today's topic:** what exactly makes them different?

Acknowledgement



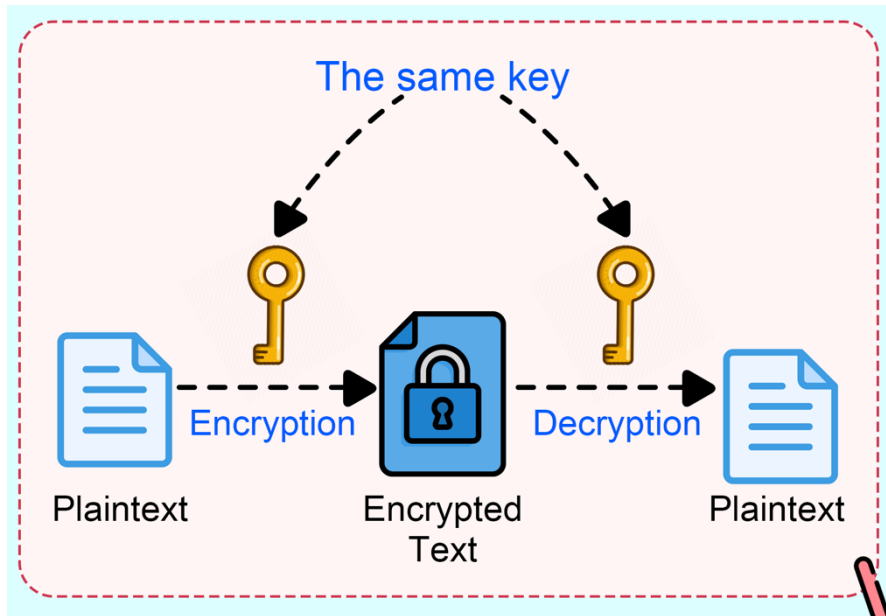
You can find it on Blackboard

The content of slides is from the textbook and Dr. Hu's supplementary materials

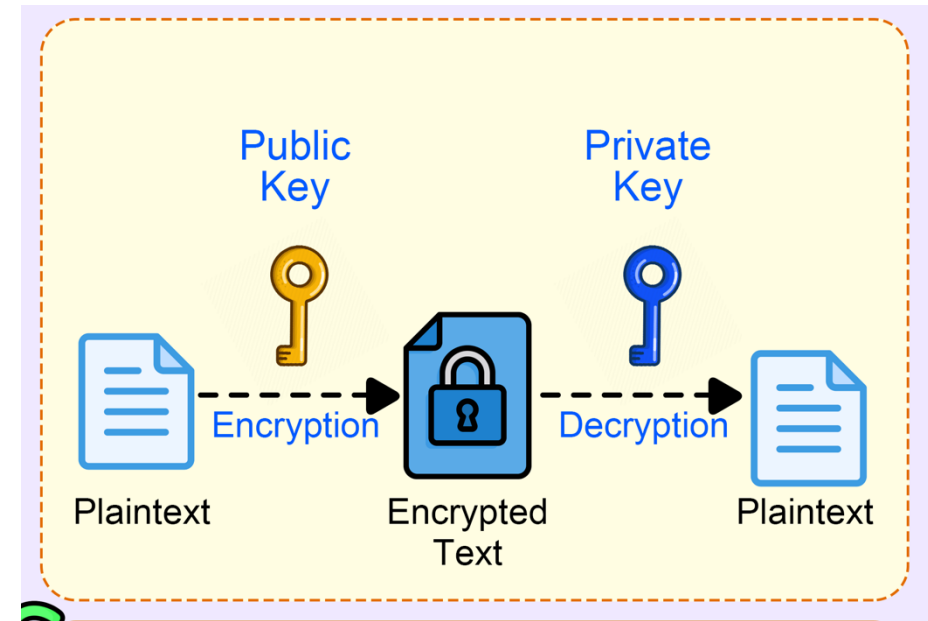
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Overview

Symmetric



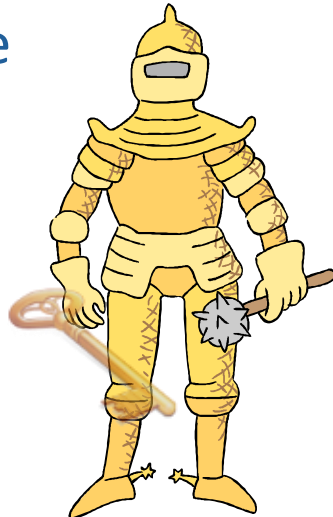
Asymmetric



Figures from: <https://bytebytego.com/guides/symmetric-encryption-vs-asymmetric-encryption/>

Symmetric Encryption

- The universal technique for providing confidentiality for transmitted or stored data
 - Also referred to as conventional encryption or single-key encryption
- Two requirements for secure use:
 - Need a strong encryption algorithm
 - Sender and receiver must have obtained copies of the secret key in a secure fashion and must keep the key secure
- Rely on bitwise operations
 - E.g., swapping, replacing, reordering



Symmetric Encryption

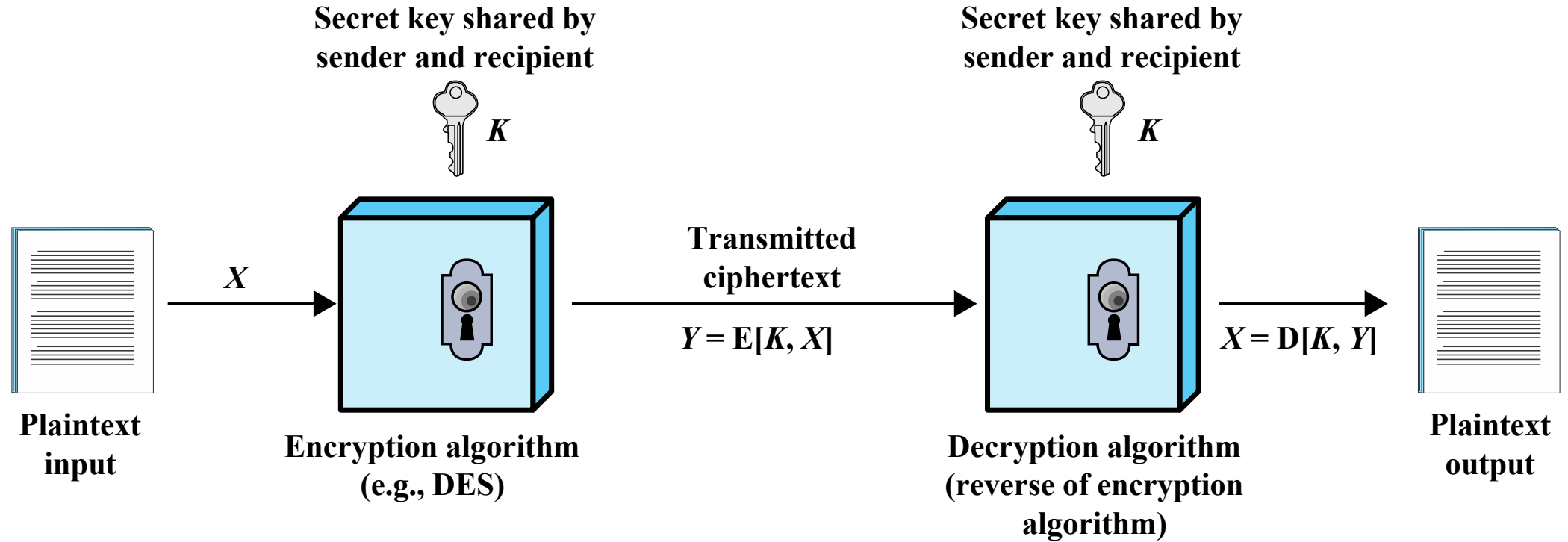


Figure 2.1 Simplified Model of Symmetric Encryption

Data Encryption Standard (DES)

- The most widely used encryption scheme
 - Adopted in 1977 by National Bureau of Standards (the predecessor of NIST)
 - Inputs 64-bit plain-text blocks and 56-bit secret key to output 64-bit cipher-text blocks
- Major concern:
 - Use of 56-bit key $\Rightarrow 2^{56} = 7.2 * 10^{16}$
 - DES has been proved insecure in 1998.
 - A DES cracker can discovered the key in 56 hours



Triple DES (3DES)

- How?
 - Repeats basic DES algorithm three times using either two or three unique 56-bit keys
 - Encryption: $C = E(K3, D(K2, E(K1, X)))$
 - Decryption: $P = D(K1, E(K2, D(K3, C)))$
- Three key length options
 - Option1: $K1 \neq K2 \neq K3$ (key length is $56 \times 3 = 168$)
 - Option 2: $K1 = K3, K1 \neq K2$ (key length is $56 \times 2 = 112$)
 - Option 3: $K1 = K2 = K3$ (key length is 56, same as DES)

Triple DES (3DES)

- Advantages:
 - 168-bit key length overcomes the vulnerability to brute-force attack of DES
 - Underlying encryption algorithm is the same as in DES
- Drawbacks:
 - Input still uses a 64-bit block size (no efficient)

Advanced Encryption Standard (AES)

- 3DES isn't considered a long-term solution
- NIST called for proposals in 1997
 - First round, 15 candidates were accepted
 - Second round, 5 candidates
- Rijndael algorithm was adopted in Nov, 2001.
 - Key length: 128, 192 and 256 bits
 - Input block size : 128 bits
 - Output block size : 128 bits

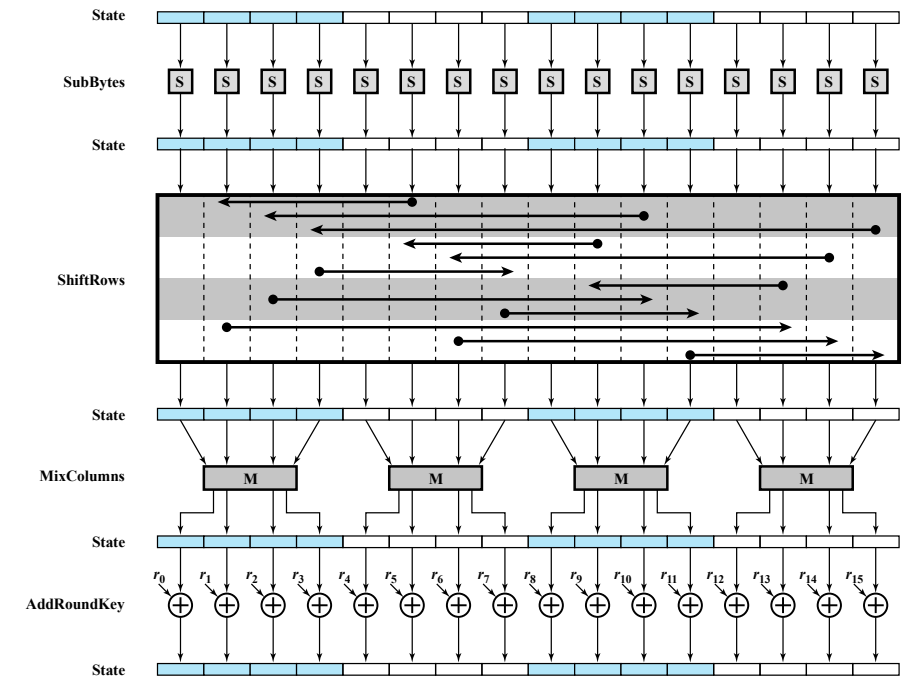


Figure 20.4 AES Encryption Round

Table: Block cipher algorithms

	DES	Triple DES	AES
Plaintext block size (bits)	64	64	128
Ciphertext block size (bits)	64	64	128
Key size (bits)	56	112 or 168	128, 192, or 256

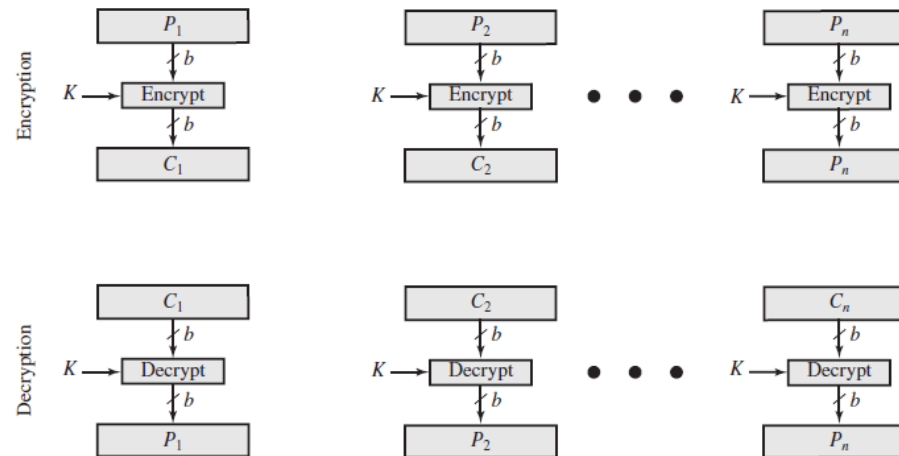
DES = Data Encryption Standard

AES = Advanced Encryption Standard

Comparison of Three Popular Symmetric Encryption Algorithms

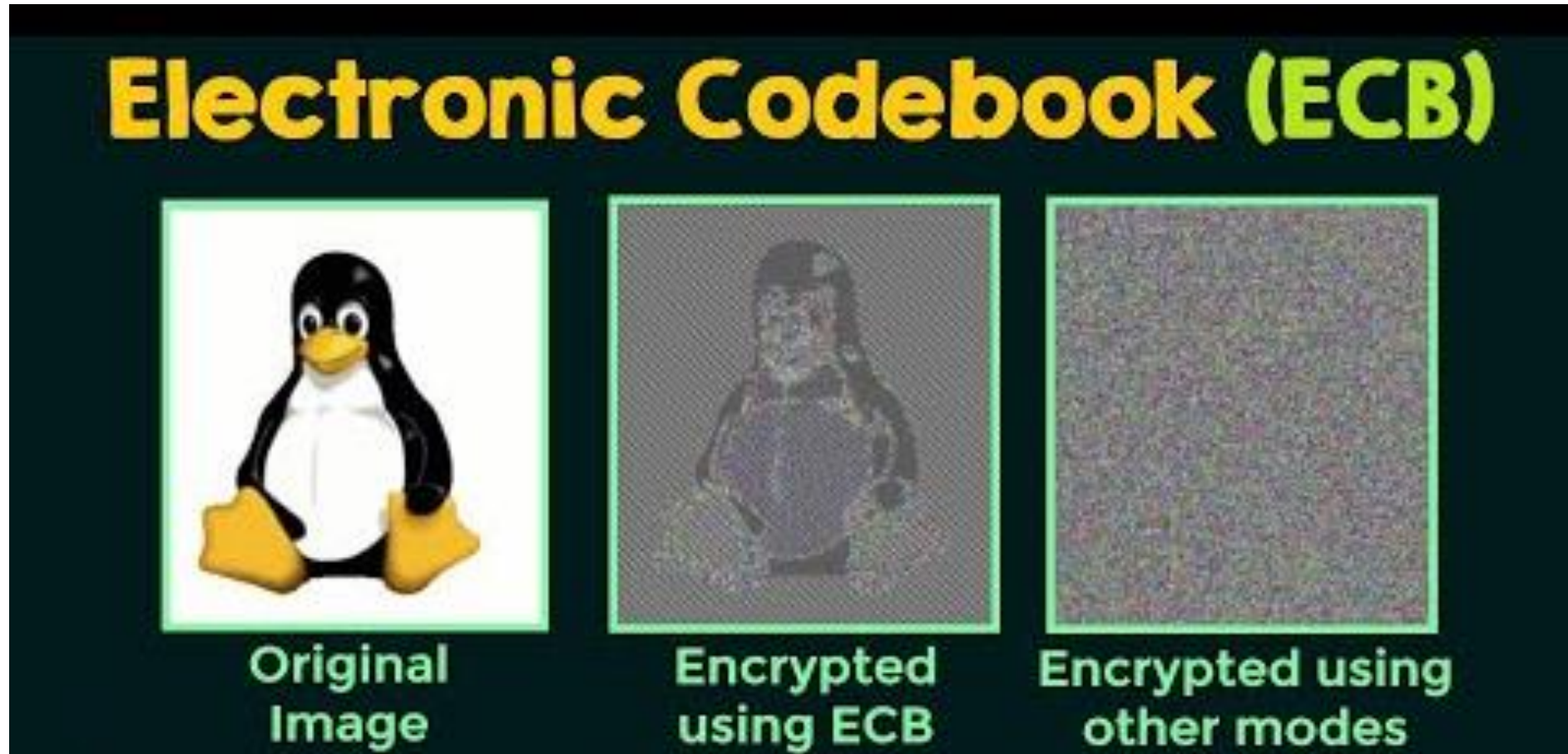
Practical security issues

- Typically, symmetric encryption is applied to a unit of data larger than a single 64-bit or 128-bit block
 - Electronic codebook (ECB) mode is the simplest approach to multiple-block encryption
 - Each block of plaintext is encrypted using the same key
 - Cryptanalysts may be able to exploit regularities in the plaintext



(a) Block cipher encryption (electronic codebook mode)

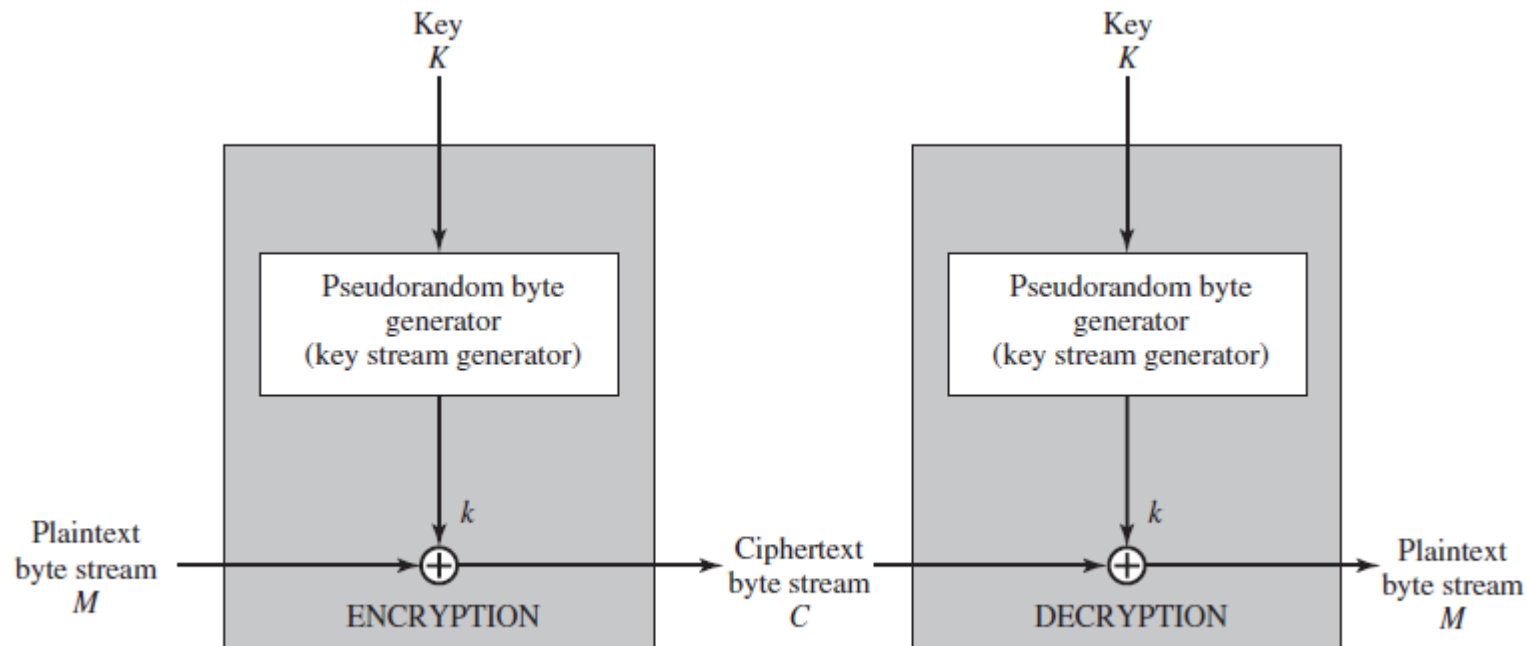
Practical security issues –the ECB Penguin



<https://www.youtube.com/watch?v=jDnenb9EHQk>

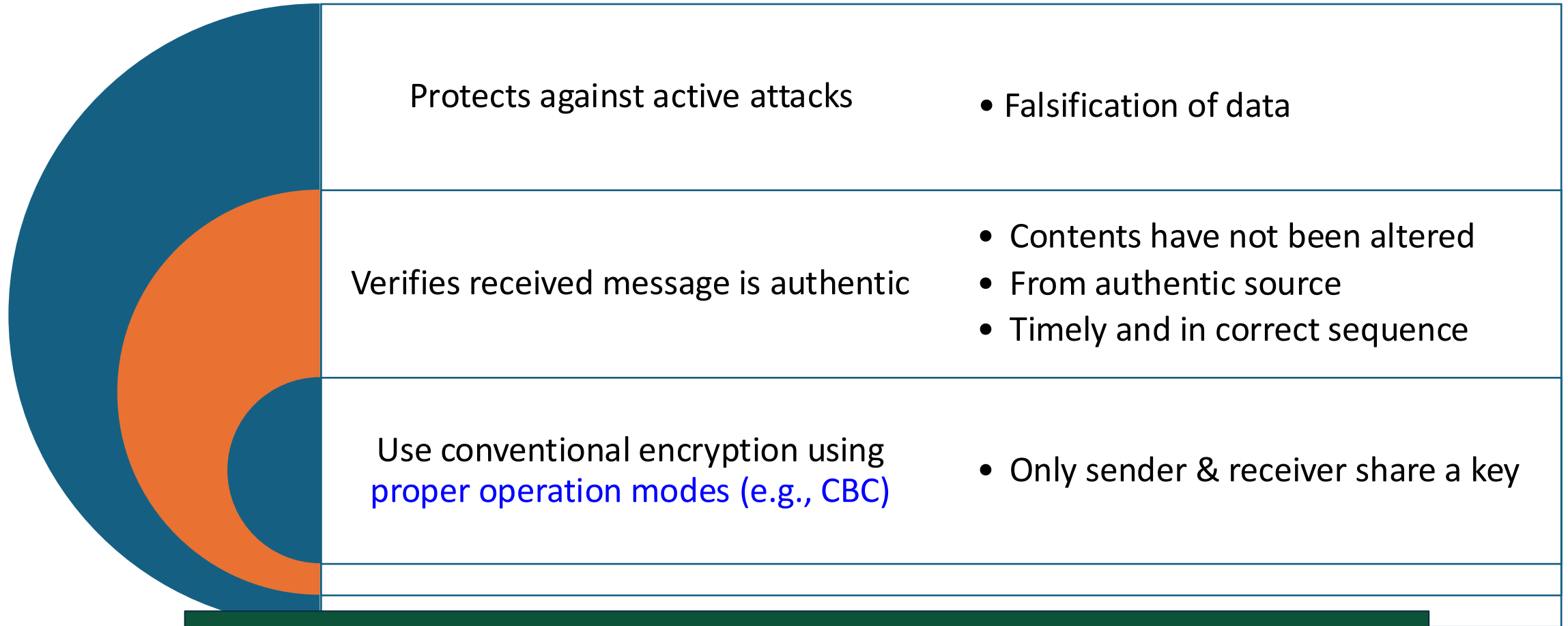
Stream cipher

- Encrypt and decrypt data byte by byte
- For each input data byte **M**, we generate one byte encryption key **K** and perform exclusive OR operation to encrypt the **M** (cipher text = **M XOR K**)



(b) Stream encryption

Message Authentication



However, encryption may not be always a good candidate.

Message Authentication Code

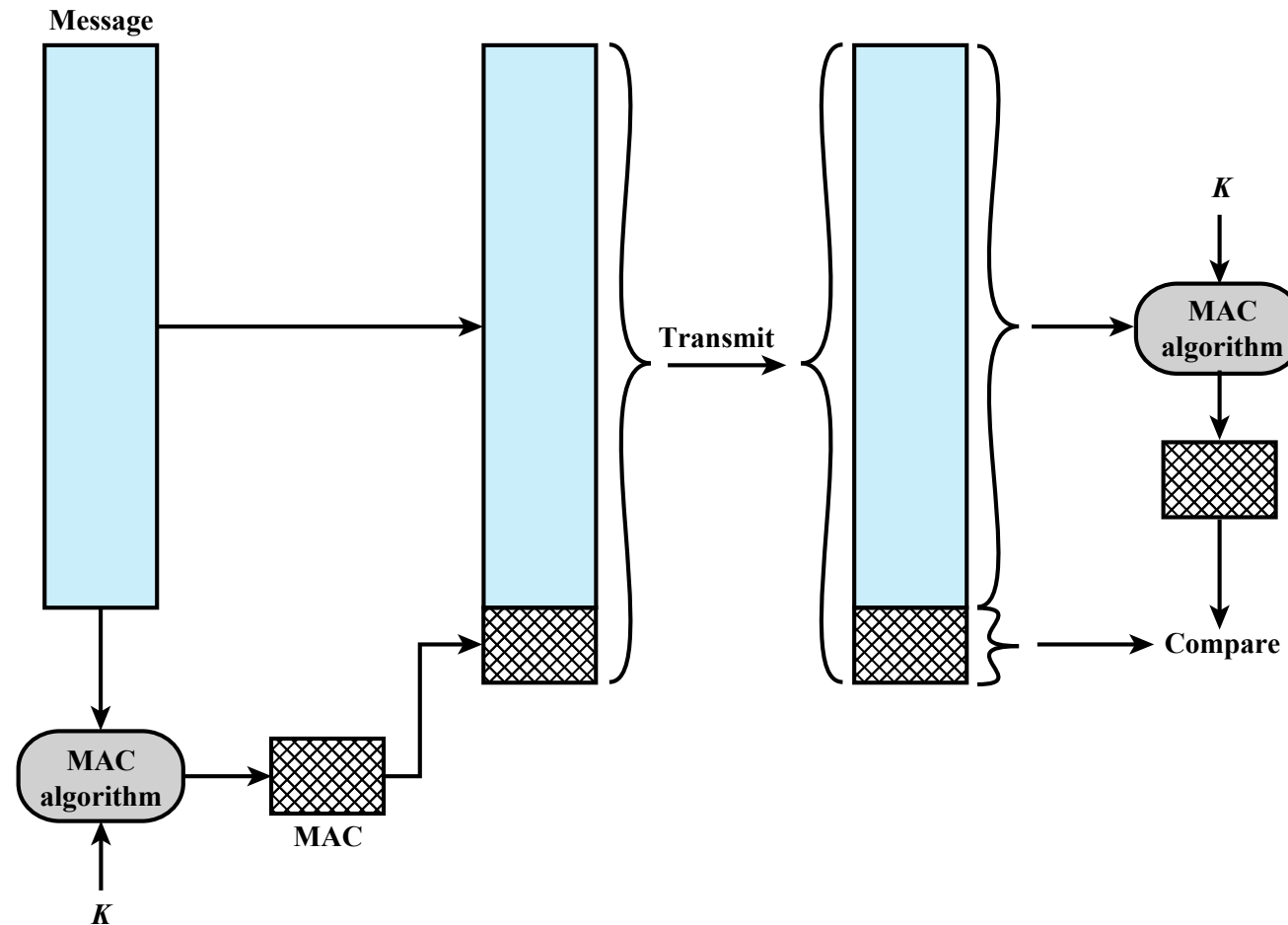


Figure 2.3 Message Authentication Using a Message Authentication Code (MAC).

Two ways to generate MAC

- Use encryption algorithms
 - E.g., using DES to encrypt a message, retrieve last 16 bits or 32 bits of ciphertext as MAC (NIST FIPS 113)
- Use hash functions
 - A hash function accepts a variable-size message and outputs a fixed-size message digest
 - MD5, SHA-1, SHA-2, SHA-3



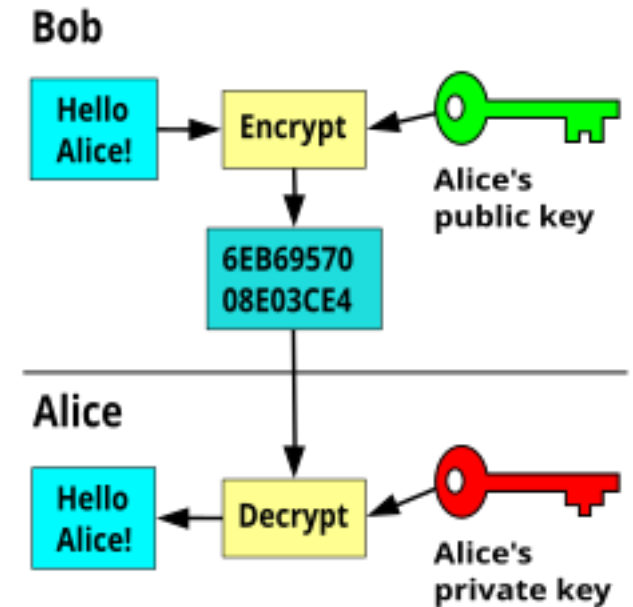
Asymmetric Cryptography

Public-Key Cryptosystems

- Proposed by Diffie and Hellman in 1976
- Based on mathematical functions (e.g., modulo and exponential operations) rather simple operations on bit patterns
 - [Diffie-Hellman key agreement \(discrete logarithm problem\)](#),
 - RSA (prime factorization problem),
 - Elliptic Curve (elliptic curve discrete logarithm problem)

Public-Key Cryptosystems

- Asymmetric
 - Use a key pair
 - A public key and a private key
 - Public key is open to the public
 - Private key is only known to the key pair owner
- If the ciphertext is encrypted by public key, it can be only decrypted by the corresponding private key
- If the ciphertext is encrypted by private key, it can be only decrypted by the corresponding public key



Public-Key Cryptosystems

- Asymmetric
 - Use a key pair
 - A public key and a private key
 - Public key is open to the public
 - Private key is only known to the key pair owner

Question: Alice wants to send a secret message to Bob. Which key does she use?

Five requirements for public-key cryptosystems

- Diffie and Hellman defined five requirements for sender A and receiver B
 - Computational easily creates a pair (public key and private key)
 - For a sender A, knowing B's public key, computational easily encrypts the message M to ciphertext C with B's public key
 - Receiver B computational easily decrypts the ciphertext C using B's private key

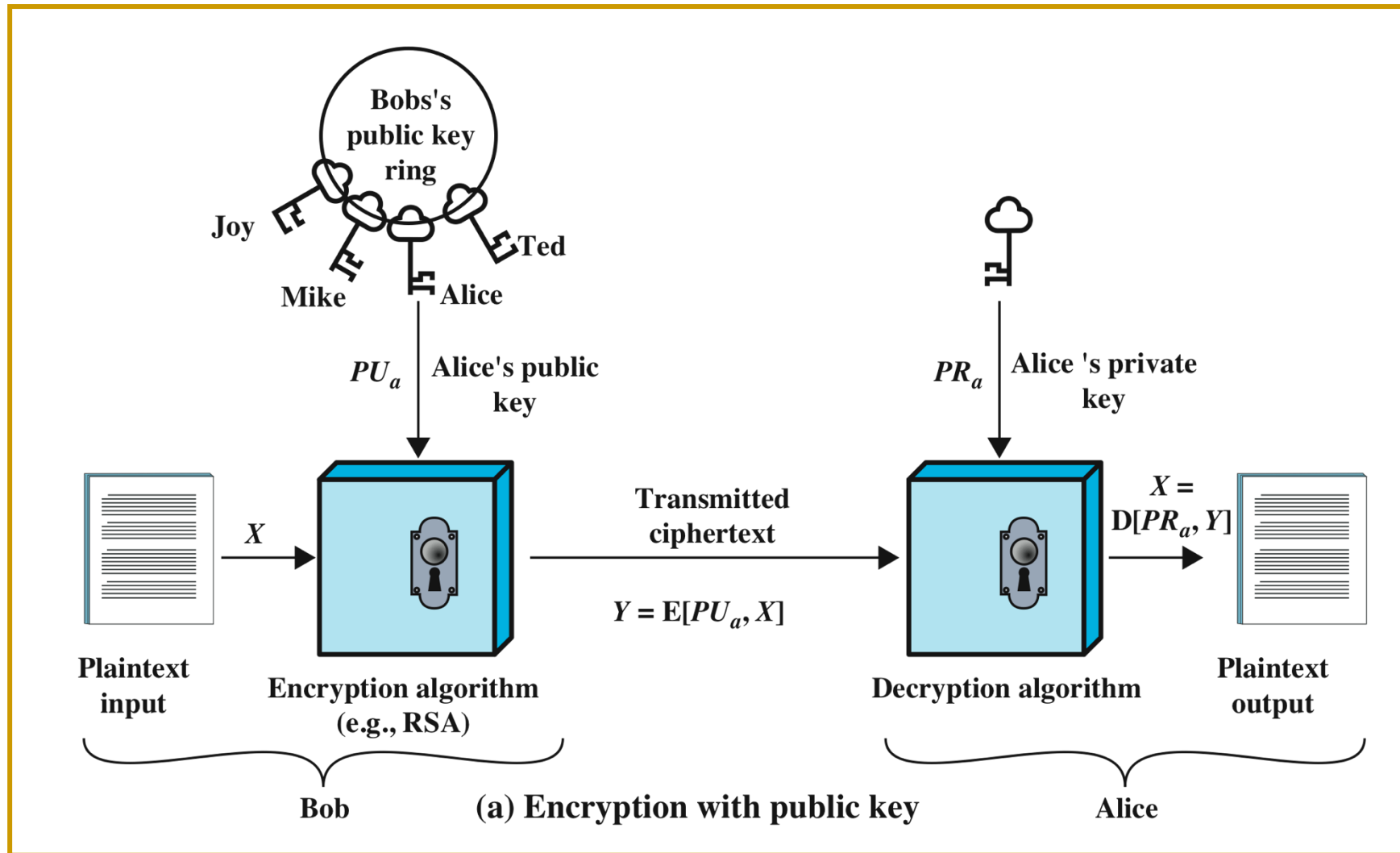
Five requirements for public-key cryptosystems

- Diffie and Hellman defined five requirements for sender A and receiver B
 - It is computational infeasible for an opponent, knowing B's public key, to determine B's private key
 - It is computational infeasible for an opponent, knowing B's public key and a ciphertext C, to recover the original message M

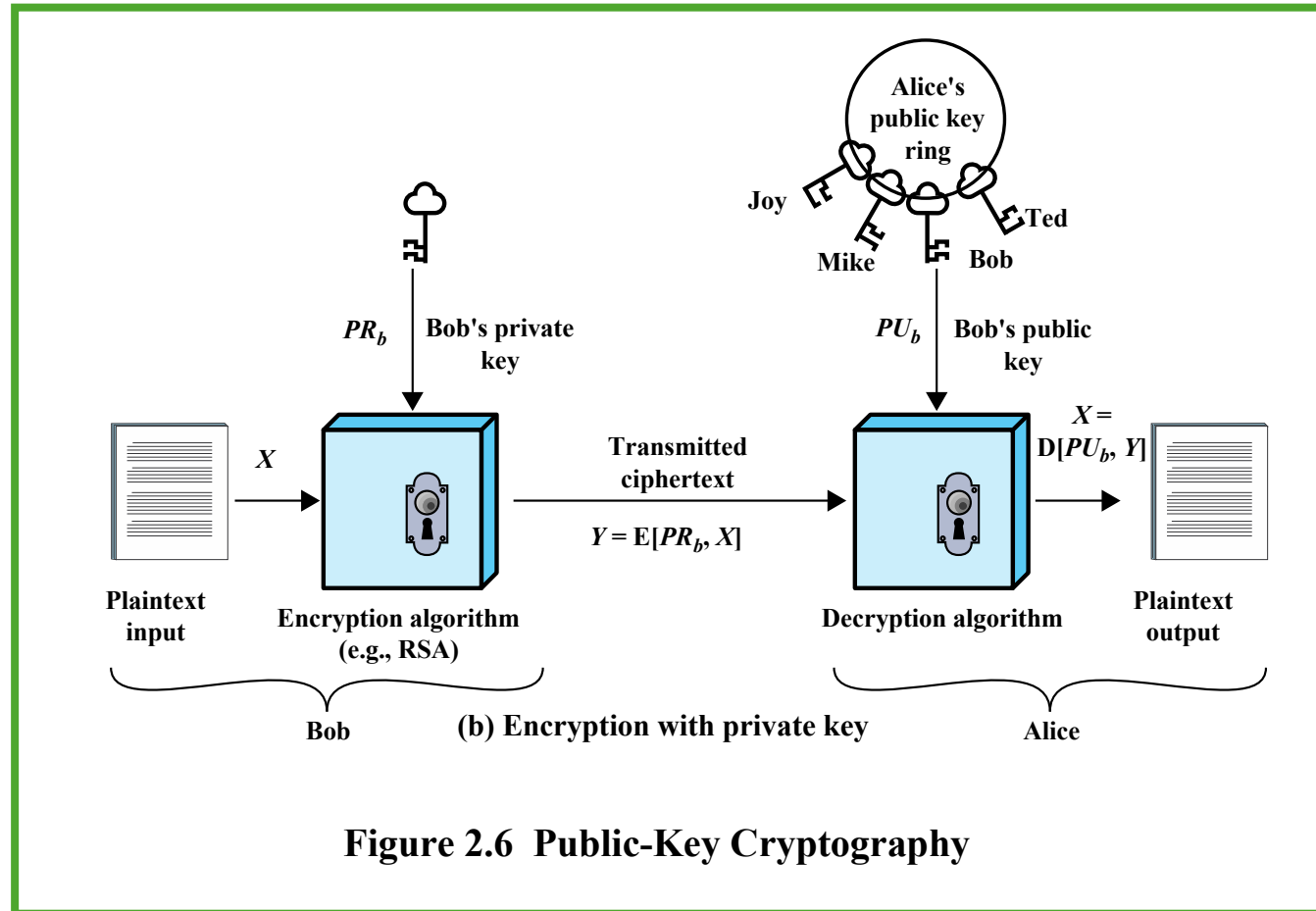
Public-key cryptosystem applications

- Data confidentiality (encryption/decryption)
- Data integrity
- Key management (Key distribution)
- Digital signature

Data confidentiality



Data integrity



- Bob encrypts data using his or her own private key
- Anyone who knows the Bob's public key will be able to decrypt the message

Why Quantum Computers Play Favorites

- **Shor's algorithm (1994):** solves prime factorization and discrete logarithms in polynomial time → breaks RSA and ECC completely, by exploiting their mathematical structure
- **Grover's algorithm (1996):** speeds up brute-force key search from 2^k to $2^{k/2}$ tries → only halves AES's effective key length, and a longer key compensates
- Shor can't touch AES: it has no algebraic structure to exploit; Grover applies to anything, but a quadratic speed-up is never fatal

That's why the world is replacing RSA/ECC while simply keeping AES with longer keys